THE MODULE ON THE SCHEME THEORETIC SUPPORT OF A COHERENT SHEAF ON A LOCALLY NOETHERIAN ALGEBRAIC SPACE IS COHERENT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The scheme theoretic support of a finite type module, and the module it carries. Morphisms of Spaces, definition spaces-morphisms-definition-scheme-theoretic-support (cited at statement line 2241). It provides the closed immersion $i : Z \to X$ and the quasi-coherent module $\mathcal{G}$ with $i_*\mathcal{G} = \mathcal{F}$, and makes the statement meaningful because a coherent module is of finite type.
- A closed immersion is locally of finite type, and a space locally of finite type over a locally Noetherian space is locally Noetherian. Morphisms of Spaces, labels spaces-morphisms-lemma-immersion-locally-finite-type and spaces-morphisms-lemma-locally-finite-type-locally-noetherian (invoked at proof lines 2251 and 2252). They show that the scheme theoretic support Z is itself locally Noetherian, which is the hypothesis under which coherence of $\mathcal{G}$ can be tested.
- Coherence on a locally Noetherian space equals finite type. Same chapter spaces-cohomology.tex, label lemma-coherent-Noetherian (invoked at proof line 2255). Since $\mathcal{G}$ is of finite type, this gives that it is coherent.

Source reference. Original lemma 07UB: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2094–2094. Navigation pointer for \ref{lemma-coherent-Noetherian} used inside the counted span (line 2255); the argument that consumes it is retained in the span.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 07U1: sources/stacks-chow/tags.txt, source lines 9956–9956. Navigation pointer for \ref{spaces-morphisms-definition-scheme-theoretic-support} used inside the counted span (line 2242); the argument that consumes it is retained in the span.

Source reference. Original lemma 06ED: sources/stacks-chow/tags.txt, source lines 8212–8212. Navigation pointer for \ref{spaces-morphisms-lemma-immersion-locally-finite-type} used inside the counted span (line 2251); the argument that consumes it is retained in the span.

Source reference. Original lemma 04ZK: sources/stacks-chow/tags.txt, source lines 6468–6468. Navigation pointer for \ref{spaces-morphisms-lemma-locally-finite-type-locally-noetherian} used inside the counted span (line 2252); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let X be a locally Noetherian algebraic space over S. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Let $i : Z \to X$ be the scheme theoretic support of $\mathcal{F}$ and $\mathcal{G}$ the quasi-coherent $\mathcal{O}_Z$-module such that $i_*\mathcal{G} = \mathcal{F}$, see Morphisms of Spaces, Definition 07U1. Then $\mathcal{G}$ is a coherent $\mathcal{O}_Z$-module.*

Proof. The statement of the lemma makes sense as a coherent module is in particular of finite type. Moreover, as $Z \to X$ is a closed immersion it is locally of finite type and hence Z is locally Noetherian, see Morphisms of Spaces, Lemmas 06ED and 04ZK. Finally, as $\mathcal{G}$ is of finite type it is a coherent $\mathcal{O}_Z$-module by Lemma 07UB □